A Low-Code Framework for Conversational AI Assistants Leveraging Retrieval-Augmented Generation and LLMs

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Abstract — Conversational AI has evolved rapidly with the advent of Large Language Models (LLMs), enabling intelligent, context-aware interactions between users and systems across various domains. These models have demonstrated remarkable capabilities in understanding natural language and generating human-like responses. However, integrating such advanced AI into enterprise applications remains a challenge, often requiring significant programming effort, AI expertise, and system customization. To address this gap, this project presents a low-code framework for building a domain-specific Conversational AI Assistant using Retrieval-Augmented Generation (RAG) and LLMs. The framework is designed to integrate seamlessly with platforms like Oracle APEX, enabling developers and business users to create intelligent assistants with minimal technical complexity. By combining the generative power of LLMs with the precision of retrieval-based methods, the assistant can deliver accurate, real-time responses sourced from enterprise knowledge bases, including documents, databases, and APIs.

Keywords — Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Low-code development, Oracle APEX, Conversational AI